Test 1 – Grade 8

Name: _____
Grade / Class: 8

Section A: English Language & Literature

1. Reading Comprehension
Read the passage below carefully and answer the questions:

"During her summer vacation, Sara decided to explore the nearby forest. She observed different plants, insects, and birds. She even recorded notes about the unique behavior of squirrels and how they stored nuts for the winter."

a) What activity did Sara do during her vacation?
b) Name two types of animals she observed.
c) Why were the squirrels storing nuts?
d) Summarize the passage in two sentences.

2. Grammar & Vocabulary
a) Convert the following sentence into passive voice:
"The scientist discovered a rare species of butterfly in the mountains."

b) Identify the adjectives in the sentence:
"The bright, colorful flowers attracted many curious bees."

c) Fill in the blank with the correct word:
"She was very _____ about presenting her project to the class." (excited / exciting)"

d) Rewrite this sentence in reported speech:
"The teacher said, 'Complete your assignments by Friday.'"

3. Writing
Write a short essay of 120–150 words on:
"The advantages and challenges of online learning for students."

Section B: Mathematics

1. Algebra
a) Solve for x: 5x−7=185x - 7 = 185x−7=18
b) Factorize: x2+7x+10x^2 + 7x + 10x2+7x+10
c) Simplify: 3a2b−5ab2+2a2b3a^2b - 5ab^2 + 2a^2b3a2b−5ab2+2a2b

2. Geometry
a) A triangle has sides of 7 cm, 24 cm, and 25 cm. Determine whether it is a right-angled triangle.
b) Find the area of a circle with radius 14 cm.

3. Ratio, Proportion & Percentage
a) A shirt originally costs $80. It is sold at a discount of 15%. Find the selling price.
b) If 3 pens cost $5, how much will 12 pens cost?

4. Data Handling
The marks of 10 students in a test are: 12, 15, 18, 20, 12, 15, 18, 20, 15, 16.
a) Find the mean, median, and mode.
b) Represent the marks in a frequency table.

Section C: Science

1. Biology
a) Name the organ responsible for regulating blood sugar levels in the human body.
b) Explain the process of photosynthesis in plants.
c) List two differences between viruses and bacteria.

2. Physics
a) A car accelerates from 0 to 60 m/s in 12 seconds. Find its acceleration.
b) Explain the difference between kinetic energy and potential energy.
c) Define refraction of light and give one example.

3. Chemistry
a) Classify the following as element, compound, or mixture:

- **Water**
- **Oxygen**
- **Salt and sand mixture**

b) Write a balanced chemical equation for the reaction between hydrochloric acid and sodium hydroxide.

4. Environmental Science
a) Explain two major causes of air pollution.
b) Give two examples of renewable energy sources.

Section D: Social Studies

1. History
a) Name two major inventions of the Industrial Revolution.
b) Explain the impact of the printing press on society.
c) Who were the major powers involved in World War I?

2. Geography
a) Define latitude and longitude.
b) Name the longest river in the world and the continent it flows through.
c) Explain two effects of deforestation on the environment.

3. Civics / Political Science
a) Explain the difference between democracy and monarchy.
b) List two fundamental rights of citizens in any democratic country.

4. Economics
a) Define supply and demand.
b) What is the difference between trade and barter?

Section E: ICT / Computer Studies

1. Computer Basics
a) Define hardware and software.
b) Give two examples of input and output devices.

2. Applications
a) Name one spreadsheet formula used to calculate the sum of numbers.
b) Explain the purpose of a presentation software.

3. Programming Basics
a) What is an algorithm? Give an example.
b) Write a simple code snippet to print "Hello, World!" in Python.

4. Cyber Awareness
a) List two rules for safe internet usage.
b) Explain what cyberbullying is.

Section F: Islamiyat

1. Quran & Hadith
a) Mention two teachings of the Quran about honesty.
b) Narrate one Hadith about the importance of helping others.

2. Aqidah / Belief
a) What are the six pillars of Iman (faith)?
b) Define Tawheed (Oneness of Allah).

3. Fiqh / Worship
a) What is the significance of Salah in Islam?
b) Name the five pillars of Islam.

4. Islamic Ethics & Morality
a) Explain the importance of respecting parents according to Islam.
b) How does Islam encourage environmental stewardship?

Section G: Arts & Music

1. Visual Arts
a) Explain the concept of perspective in drawing.
b) Name two primary colors.

2. Music
a) Define rhythm and melody.
b) Give an example of a traditional musical instrument from any culture.

Section H: Health & Physical Education

1. Physical Fitness
a) Name two exercises to improve flexibility.
b) List one team sport and one individual sport.

2. Health & Hygiene
a) Why is a balanced diet important?
b) Mention two ways to maintain personal hygiene.

3. Safety & Environment
a) Name one precaution to take during an earthquake.
b) Give one example of how to conserve water at home.